

Paper Title	Published Where	Authors	Abstract			Link of the Paper
Chaotic Time Series Analysis	ResearchGate	Zonghua Liu	chaotic dynamical systems are ubiquitous in nature and most of them does not have an explicit dynamical equation and can be only understood through the available time series. We here briefly review the basic concepts of time series and its analytic tools, such as dimension, Lyapunov exponent, Hilbert transform, and attractor reconstruction. Then we			https://www.researchgate.net/publication/43170506_Chaotic_Time_Series_Analysis
Survey:Time-series data preprocessing: A survey and an empirical analysis	Journal of Engineering Research	Amal Tawakuli a. *, Bastian Havers b, Vincenzo Guisano c, Daniel Kaiser a, Thomas Engel	Data are naturally collected in their raw state and must undergo a series of preprocessing steps to obtain data in their input state for Artificial Intelligence (AI) and other applications. The data preprocessing phase is not only necessary to fit input requirements but also effective in improving AI training efficiency and output accuracy. Data preprocessing is a time consuming and complex phase that lacks a unified and structured approach. We survey data			https://arxiv.org/abs/2406.14991
DChaos: An R Package for Chaotic Time Series Analysis	The R Journal	Julio E. Sandubete and Lorenzo Esco	Chaos theory has been hailed as a revolution of thoughts and attracting ever-increasing attention of many scientists from diverse disciplines. Chaotic systems are non-linear deterministic dynamic systems which can behave like an erratic and apparently random motion. A relevant field inside chaos theory is the detection of chaotic behavior from			https://arxiv.org/abs/2510.13835
The analysis of chaotic time series	ResearchGate	Joshua D. Reiss	Chaotic time series analysis methods were applied to several experimental systems. Analysis of a Poincare section of magnetoelastic ribbon time series was used to construct symbolic dynamics and extract empirical quantities such as fractal dimension and Lyapunov exponents. In the pulse thermal combustion engine, analysis of several data sets was used to			https://arxiv.org/abs/2502.13897
Chaotic time series prediction of nonlinear systems based on various neural network models	Science Direct	Ying Sun a, Luying Zhang a, Minghui Yao b Show more	This paper discusses the chaos prediction of nonlinear systems using various neural networks based on the modified substructure data-driven modeling architecture. In the modeling step, we construct two-coefficient loss functions according to the linear multi-step method to improve the prediction accuracy of neural networks. Then, the predicted response data of the system is given by the forward Euler method and neural networks. Under such architecture, chaos			https://arxiv.org/abs/2410.07331